

Loops in Python

Programs that contain repeating lines of code can be time-consuming to type in and difficult to understand. A clearer way of writing them is by using a loop command. The simplest loops are ones that repeat a certain number of times, such as “for” loops.

SEE ALSO

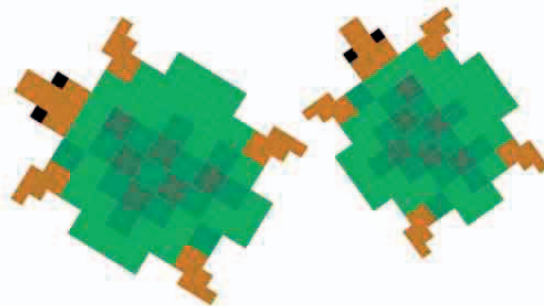
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loops

Repeating things

A “for” loop repeats the code without having to type it in again. It can be used to repeat something a certain number of times, for example, if you want to print the names of a class of 30 students.



1 Program the turtle

A “for” loop can also be used to shorten the code. This program allows the user to control a turtle that draws a line as it moves around the screen. The user can draw shapes on the screen, such as a triangle, by directing the turtle’s movements.



This makes the turtle turn 120 degrees to the right

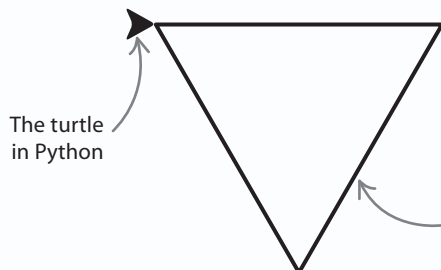
```
from turtle import *
forward(100)
right(120)
forward(100)
right(120)
forward(100)
right(120)
```

Loads all the commands that control the turtle

This command moves the turtle forward

2 The turtle draws a triangle

The program tells the turtle how to draw a triangle by giving it the length of the three sides and the angles between them. The turtle will appear in a separate window when you run the program.



The turtle in Python

The program makes the turtle draw a triangle

3 Use a “for” loop

The program above gives the turtle the same two commands, “forward(100)” and “right(120)”, three times – once for each side of the triangle. An alternative to this is to use these two commands inside a “for” loop. Try drawing a triangle simply using the code shown below.

```
for i in range(3):
    forward(100)
    right(120)
```

The “for” loop tells the program to repeat the instructions three times

The block of instructions in a loop is indented by four spaces